Time: 1 hr. Max. Marks: 12

Paper Code: DSC–503

B. Tech. CSE-Data Science 5th Semester Examination Session 2026-27

Note: Attempt All Question

First Mid Semester Examination

Computer Networks

Q1. Differentiate the following with respect to OSI layers functionality: [4]
CO1
(a) Logical Address, [BL-2]
(b) Physical Address
(c) Port Address

Q2. A sender needs to transmit the data contained in the 8-character text "Binomial." The text is divided into 2-byte (16-bit) words. Each character is represented using ASCII, and each ASCII byte is expressed as a two-digit hexadecimal value. [4] [BL-3,5]
CO2
(a) Calculate the partial-sum and checksum at the sender site and at the receiver site.
(b) Suppose that the character 'B' in the first data word is changed to 'D' during transmission. At the receiver, calculate the resulting checksum and determine whether the receiver can detect the transmission error.

Q3. Using 6-bit sequence numbers, what is the maximum size of the send and receive windows for each of the following protocols? [4] [BL-3,5]
CO2
(a) Stop-and-Wait ARQ
(b) Go-Back-NARQ
(c) Selective-Repeat ARQ

What are the advantages & disadvantages of Go-Back-N ARQ compared to Selective Repeat ARQ.